Computer Science & IT

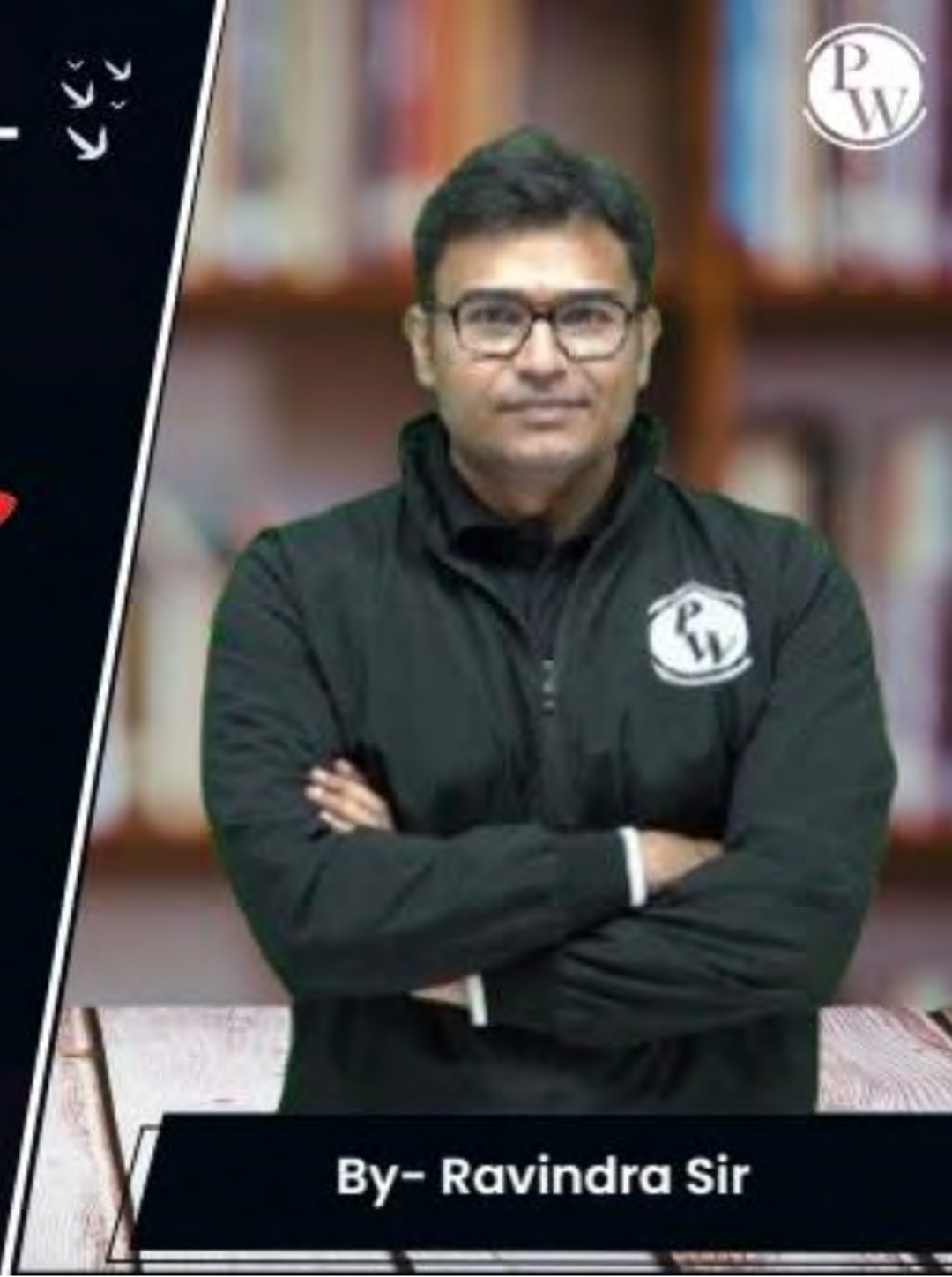
Theory of Computation





Regular Languages

Lecture No. 15



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Regular

Topics to be Covered



- Topic
- Topic
- Topic

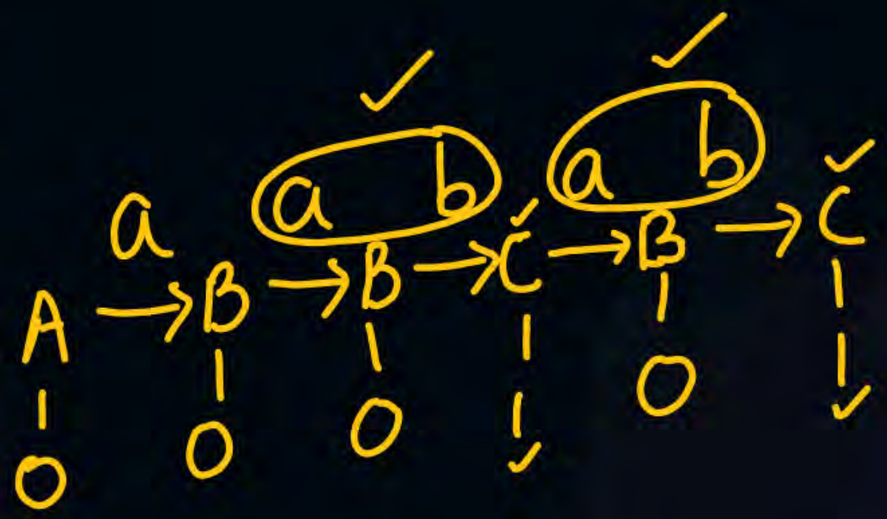
Regular, Moore, Mealy

Construct a mode machine that takes set of all strings over {a,b} and prints '1' as o/p for every occurance of ab as a substring mode mc → o/p is associated with the state

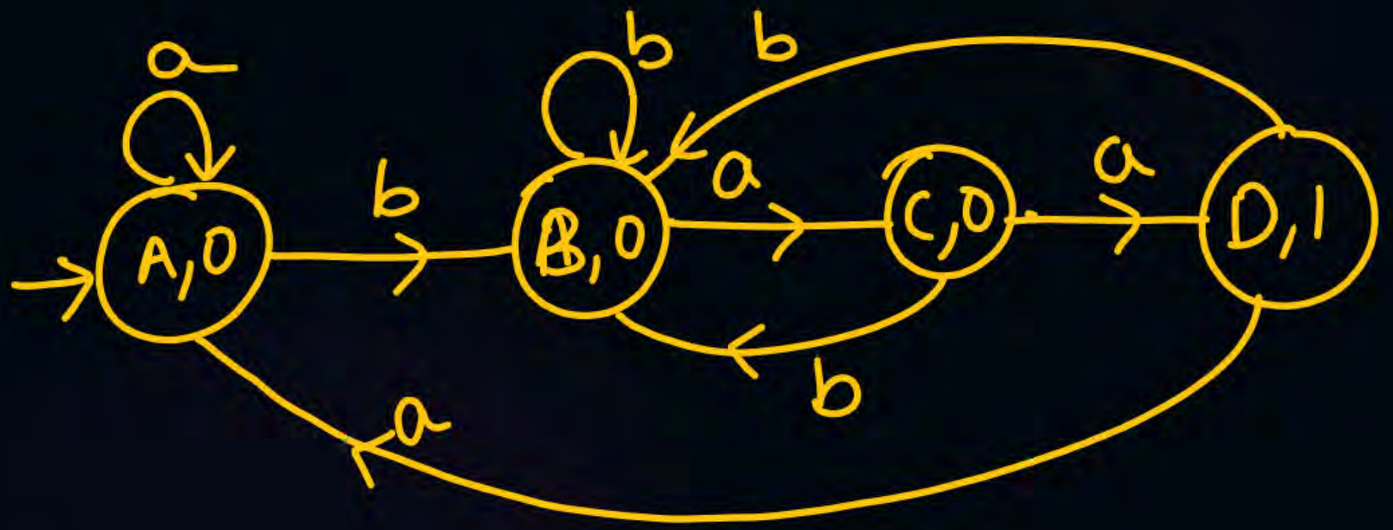


i/p → n

o/p → (n+1)

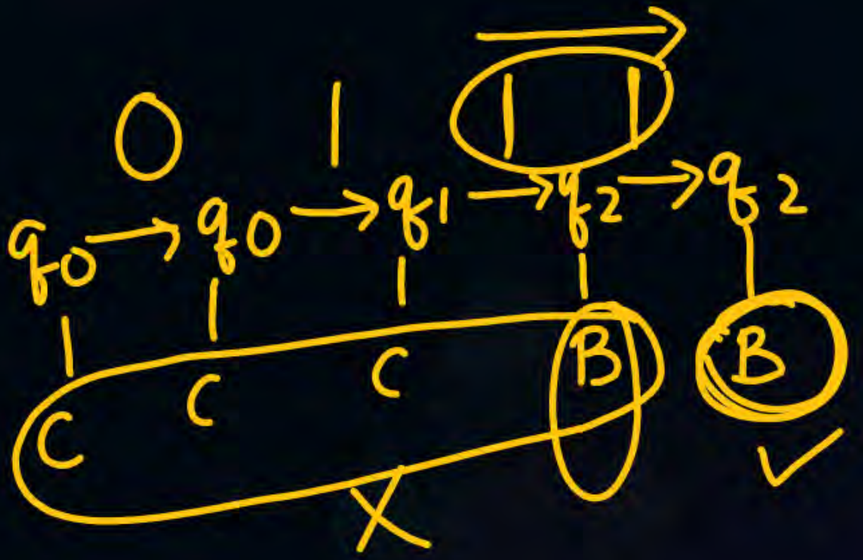


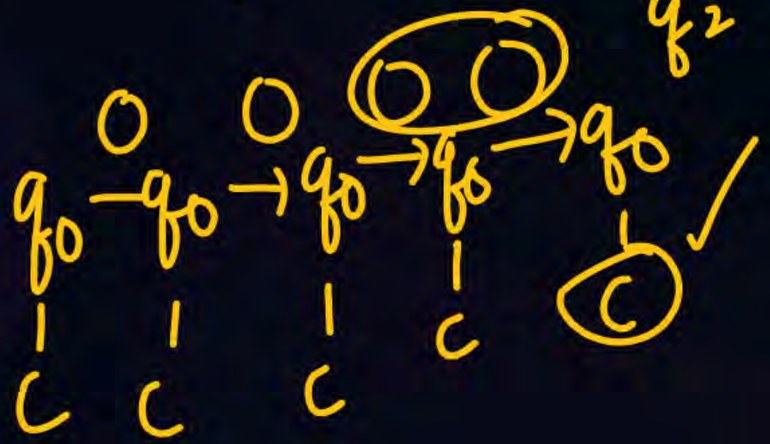
Construct a mode m/c, that prints '1' fd every occurance of the 'baa'



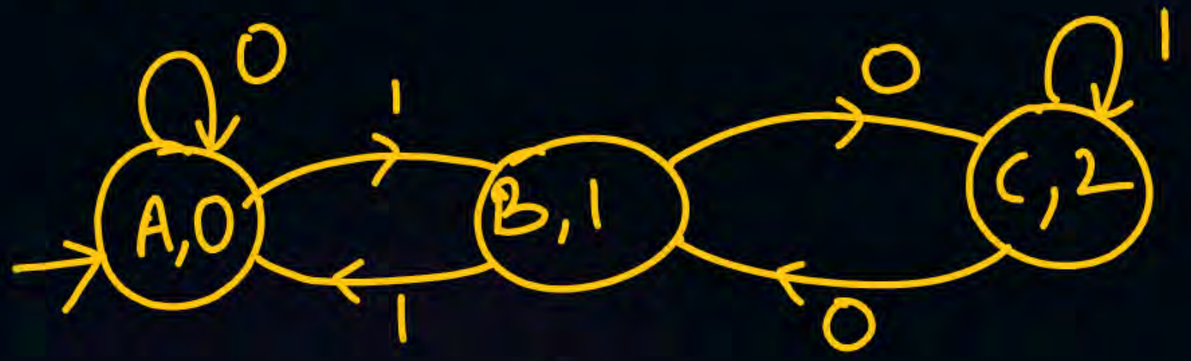
→ Construct a mode m/c that takes set of all strings over {0,1} and produces 'A' as o/p if i/p ends with '10' or produces 'B' as o/p if i/p ends with '11' & produces 'C' otherwise







→ Construct a mode m/c that takes binary no as i/p and produces residue modulo 3 as o/p



	0	1	
A	A	B	0
B	C	A	1
C	B	C	2

→ Construct a mode m/c that takes base 4 number as i/p and produces residue modulo 5 as o/p

	0	1	2	3	
→ q_0	q_0	q_1	q_2	q_3	0
q_1	q_4	q_0	q_1	q_2	1
q_2	q_3	q_4	q_0	q_1	2
q_3	q_2	q_3	q_4	q_0	3
q_4	q_1	q_2	q_3	q_4	4

mealy m/c :-

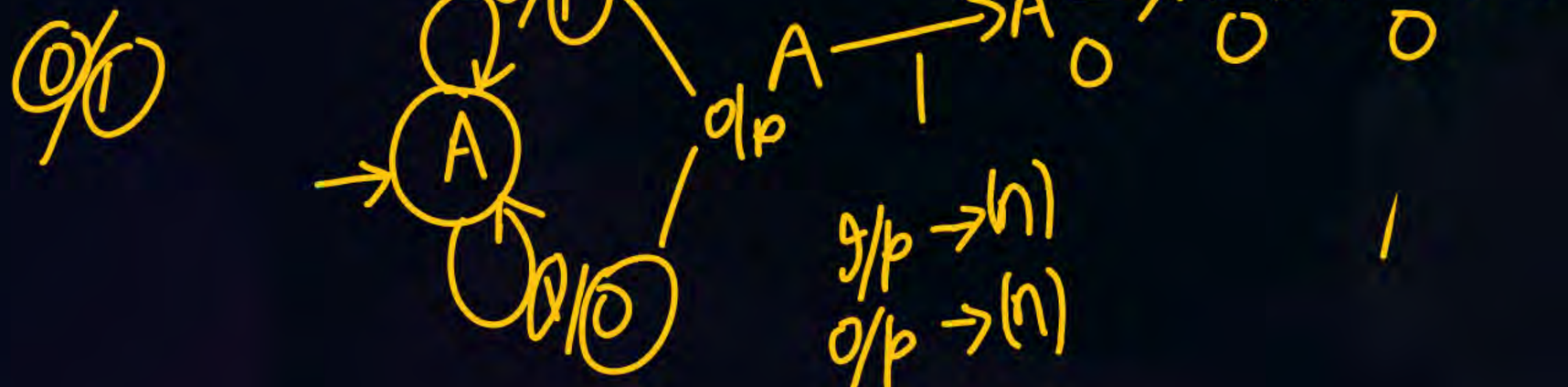
Construct a mealy m/c that takes binary number as i/p and produces 1's complement as o/p

mealy → o/p is associated with transitions

mode → " " " states

1 0 1 1

0 1 0 0



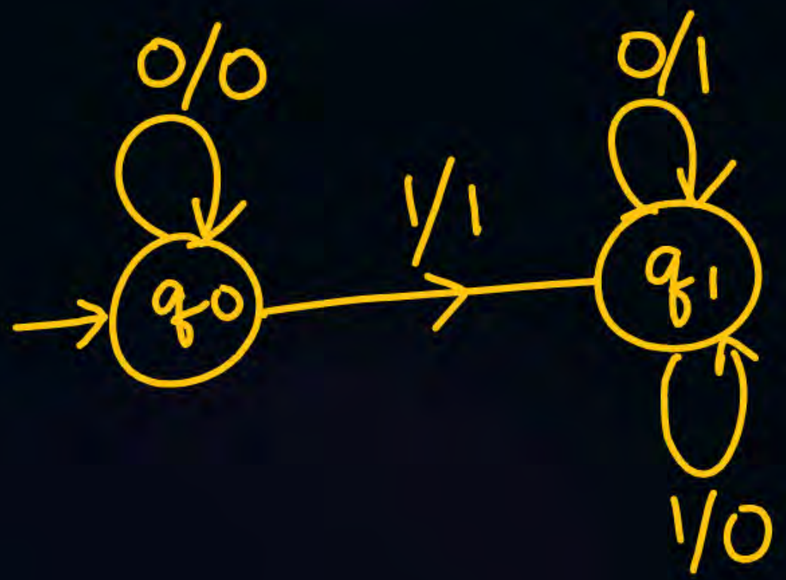
o/p → (1)

o/p → (1)

1

→ Construct a mealy m/c for 2's complement, assuming that i/p is read from LSB to MSB and carry is discarded.




0/0
0/1
1/1
q0
q1
1/0

THANK - YOU